

SMART HOME ELECTRICITY MONITORING & CONTROL SYSTEM

A PROJECT REPORT

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in the partial fulfillment for the award of the degree of

BACHELOR OF TECHNOLOGY

in

ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

JEPPIAAR INSTITUTE OF TECHNOLOGY, SRIPERUMBUDUR

(An Autonomous Institution)

ANNA UNIVERSITY: CHENNAI 600025

APRIL 2026

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BONAFIDE CERTIFICATE

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This project report is submitted for viva voce examination to be held on

INTERNAL EXAMINER

EXTERNAL EXAMINER

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ACKNOWLEDGEMENT

We are personally indebted to a number of persons that a complete acknowledgement would be encyclopedic. First, I love to record my deepest gratitude to the Almighty Lord and my family.

We would like to express our deepest gratitude and respect to our Honorable **Col. Dr. JEPPIAAR, M.A., B.L., Ph.D., Chairman** for having given the opportunity to pursue the education in this premier institution

Our sincere thanks and performed sense of gratitude goes to the respected chairman for all his effort in educating me in a premier institution.

We take this opportunity to thank our Managing Director of this prestigious institution **Dr. N. MARIE WILSON, B.Tech. MBA., Ph.D.,** Jeppiaar Institute of Technology, for providing facilities in the college premises for carrying out this project work.

We express our gratitude to our principal **Dr. J. VENU GOPALA KRISHNAN, B.E, M.E, Ph.D.,** and **Mr. S. Saravanan, M.E, Ph.D.** the Head of the Department, of Artificial Intelligence and Data Science, for their guidance and advise all through the project.

We convey our gratitude to our internal guide **Mr.D.Shanmugavel**,for his valuable guidance throughout the duration of this project. We would also like to thank our parents and friends for the support they extended during this course of the project.

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ABSTRACT

The rapid advancement of smart home technologies has created an increasing demand for energy-efficient, safe, and automated residential systems. This project presents a Smart Home Electricity Monitoring and Control System designed to optimize energy consumption while enhancing safety and user convenience. The system employs the ESP32 microcontroller as the central processing unit to monitor voltage, current, and power parameters in real time using dedicated sensors. Household electrical appliances, including lights, fans, and other devices, are controlled through relay modules, enabling automated scheduling and remote operation via a mobile application. The LCD interface displays live electrical data, allowing users to observe consumption patterns directly, while a buzzer alert system notifies users of abnormal conditions such as overcurrent or voltage fluctuations, preventing potential hazards. Integration with IoT technology facilitates real-time data transmission and remote management, empowering users to make informed decisions regarding energy use. The system also incorporates intelligent algorithms to identify power wastage and optimize appliance control, promoting energy conservation and cost reduction. By combining real-time sensing, wireless communication, and automated control, the proposed solution offers a scalable, reliable, and user-friendly approach to modern home energy management. This system not only ensures enhanced safety and convenience but also contributes to sustainable energy practices. The system also incorporates a machine learning-based prediction model to estimate the monthly electricity cost based on usage patterns. The project demonstrates that leveraging IoT-based monitoring and control in residential environments can significantly improve energy efficiency and provide a practical, modern solution for smart home applications.

Keywords: Smart Home, Electricity Monitoring, ESP32, IoT, Energy Efficiency, Automated Control, Electrical Safety, Remote Monitoring.

LIST OF ABBREVIATIONS

IoT – Internet of Things

ESP32 – WiFi enabled microcontroller

LCD – Liquid Crystal Display

AC – Alternating Current

DC – Direct Current

API – Application Programming Interface

GPIO – General Purpose Input Output

MQTT – Message Queue Telemetry Transport

CHAPTER 1

1.AIM:

The primary aim of the **Smart Home Electricity Monitoring and Control System** is to design and implement an intelligent, automated solution that enhances energy efficiency, safety, and convenience within residential environments. Modern households face challenges such as unmonitored electricity consumption, high energy costs, potential electrical hazards, and the manual operation of multiple appliances. This project aims to address these issues by integrating advanced sensing, automated control, and IoT-based remote monitoring into a unified system.

Specifically, the system seeks to continuously monitor key electrical parameters, including voltage, current, and power consumption, using dedicated sensors connected to the **ESP32 microcontroller**. Real-time data is displayed on an **LCD screen**, allowing users to observe consumption patterns and identify energy-intensive devices. The system also incorporates relay-based control mechanisms to automate the operation of appliances such as lights, fans, and other household devices, reducing manual intervention and preventing unnecessary energy wastage.

A critical component of the aim is to improve electrical safety by detecting abnormal conditions, including overcurrent, voltage fluctuations, and potential short circuits. Alerts are generated through a **buzzer** and transmitted via **IoT connectivity** to a mobile application for immediate user notification. Ultimately, the project aims to provide a scalable, reliable, and user-friendly smart home solution that promotes sustainable energy usage, reduces electricity costs, enhances safety, and delivers modern convenience to residents.

1.1.INTRODUCTION:

The concept of smart homes has transformed the way residential environments are managed, integrating technology to provide convenience, efficiency, and safety. In contemporary society, households face challenges related to increasing energy consumption, electrical hazards, and manual operation of multiple appliances. Traditional electrical systems are limited in providing real-time monitoring, automated control, or remote management, often leading to unnecessary power wastage, increased energy bills, and safety risks. With growing environmental awareness and the need for sustainable energy practices, developing systems that combine monitoring, control, and intelligent management of household electricity has become imperative.

The Smart Home Electricity Monitoring and Control System addresses these challenges by employing the ESP32 microcontroller, a high-performance, low-power, and Wi-Fi-enabled platform, capable of interfacing with sensors and relay modules for real-time monitoring and control. The system continuously measures electrical parameters such as voltage, current, and power consumption using precise sensors. The collected data is displayed on an LCD screen, allowing users to directly observe the operational status and energy usage of connected appliances. Abnormal conditions, such as overcurrent, short circuits, or voltage fluctuations, are immediately identified, and a buzzer alert notifies users to prevent potential hazards.

An essential feature of this system is IoT integration, which allows remote access and control of appliances via a mobile application. Users can switch devices on or off, schedule operations, and monitor energy consumption patterns from anywhere, enhancing convenience and promoting energy-efficient behavior. Intelligent algorithms analyze usage trends to identify appliances consuming excess power, helping households optimize electricity consumption.

This system is designed to be scalable and adaptable, allowing future integration with additional smart home devices such as air conditioners, smart meters, and renewable energy sources. By combining real-time sensing, wireless communication, and automated

control, the proposed system improves energy management, ensures electrical safety, and reduces human dependency on manual monitoring. It represents a practical solution aligned with modern smart home applications, addressing both sustainability and user convenience. Furthermore, the project emphasizes the importance of smart technologies in fostering energy awareness, promoting responsible usage, and creating a safer and more efficient living environment.

In conclusion, the Smart Home Electricity Monitoring and Control System demonstrates that integrating advanced microcontrollers, sensor technologies, and IoT capabilities can transform traditional residential electrical systems into intelligent, automated, and safe networks, paving the way for modern energy-efficient homes.

1.2 OBJECTIVES

The primary objective of this project is to design and implement a Smart Home Electricity Monitoring and Control System that enhances energy efficiency, electrical safety, and user convenience within residential environments. Traditional electrical systems are largely passive, providing no feedback on energy consumption or appliance performance. As a result, users are unable to monitor real-time electricity usage, identify abnormal conditions, or control appliances remotely. The proposed system aims to address these limitations by providing a comprehensive solution that integrates real-time monitoring, automated control, and IoT-based communication.

The first objective is to enable accurate and continuous monitoring of electrical parameters. By utilizing voltage and current sensors, the system measures power consumption for each connected appliance. This data is processed using the ESP32 microcontroller, which calculates real-time energy usage and detects abnormal fluctuations. Monitoring these parameters allows users to make informed decisions regarding appliance operation, reduce power wastage, and identify energy-intensive devices.

The second objective is to implement automated control of household appliances through relay modules. The system can switch devices on or off based on pre-programmed schedules, sensor input, or remote commands from a mobile application. Automation ensures that appliances are operated only when needed, contributing to energy savings and reducing unnecessary manual intervention.

The third objective is to ensure electrical safety within the home environment. Abnormal conditions, including overcurrent, voltage spikes, or potential short circuits, are detected immediately. Alerts are generated through a buzzer, and critical information is transmitted to the mobile application, enabling timely corrective action. By integrating intelligent safety measures, the system minimizes the risk of fire, damage to electrical appliances, and potential harm to residents.

The fourth objective is to enable remote monitoring and management using IoT technology. The system sends live data to a mobile application, allowing users to track electricity consumption, control appliances, and receive alerts from any location. Remote accessibility enhances convenience, supports energy-conscious behavior, and allows households to respond to emergencies without physical presence.

Additional objectives include providing a user-friendly interface, scalability for future expansion, and integration with other smart home devices such as smart meters and renewable energy systems. The system is also designed to support data analytics, enabling users to analyze trends and optimize energy usage further.

In summary, the objectives of the proposed system focus on improving energy efficiency, ensuring electrical safety, facilitating automation, and providing remote management capabilities. Through precise monitoring, intelligent control, and IoT integration, the project aims to create a modern, sustainable, and practical solution for smart home environments, promoting responsible energy consumption and convenience for users.

1.3 PROBLEM STATEMENT

Residential electricity management faces significant challenges due to the lack of real-time monitoring, inefficient energy usage, and potential safety hazards. In traditional homes, electricity consumption is largely unmonitored, and appliances are operated manually, often leading to unnecessary energy wastage, higher electricity bills, and increased environmental impact. Users have limited visibility into the consumption patterns of individual appliances, making it difficult to identify energy-intensive devices or optimize usage for cost efficiency.

Another major concern is electrical safety. Overloaded circuits, voltage fluctuations, short circuits, and appliance malfunctions can result in accidents, equipment damage, or even fire hazards. Conventional systems rarely provide timely alerts, and users may remain unaware of developing issues until significant damage occurs. This lack of monitoring and proactive control increases the risk to both people and property.

Manual operation of multiple appliances further complicates energy management. Users may forget to switch off devices, leading to continuous energy consumption even when appliances are idle. Additionally, households with elderly members, physically disabled individuals, or occupants with busy schedules may find it challenging to manage appliances efficiently. This highlights the need for a system that combines automation with intelligent monitoring.

While some commercial solutions exist, many are limited in scope, costly, or lack integration with IoT-based remote monitoring. Existing systems often fail to provide real-time alerts, comprehensive energy analytics, or user-friendly interfaces that promote energy-conscious behavior. There is a gap for an affordable, scalable, and reliable solution that can monitor, control, and manage household electricity intelligently.

The proposed problem can be summarized as the need to develop an integrated system that continuously monitors electrical parameters, detects anomalies, enables automated

control, and allows remote access through IoT technology. Such a system should address energy wastage, enhance safety, and improve user convenience while being adaptable to modern smart home requirements.

In conclusion, the absence of real-time electricity monitoring, automated appliance control, and IoT-enabled remote management presents a critical challenge in modern homes. Developing a Smart Home Electricity Monitoring and Control System can bridge this gap by providing precise monitoring, intelligent control, safety alerts, and remote accessibility. This solution will empower users to optimize energy consumption, reduce electricity costs, prevent hazards, and create a safer, more efficient living environment. The system also serves as a foundation for future expansion, integrating renewable energy sources, smart meters, and other smart home devices for a fully connected and sustainable residential ecosystem.

CHAPTER 2

2.SYSTEM ANALYSIS

EXISTING SYSTEM

Traditional home electricity management relies primarily on manual monitoring and operation of appliances. In most households, electrical devices such as lights, fans, air conditioners, and kitchen appliances are turned on or off manually without any automated scheduling or centralized monitoring system. Users typically depend on physical switches, timers, or rudimentary energy meters to track electricity consumption. While analog or digital meters may record total power usage over time, they provide no real-time feedback on individual appliances or instantaneous electrical parameters such as voltage, current, or power factor.

Existing systems lack the integration of intelligent controllers or automation technologies, which limits energy optimization. In conventional setups, appliances may remain

operational even when not required, contributing to unnecessary energy consumption. Moreover, traditional systems do not provide safety mechanisms to alert users about abnormal electrical conditions such as overcurrent, voltage spikes, or short circuits. As a result, households face increased risk of electrical hazards, equipment damage, and potential fire incidents.

While some commercially available smart plugs or standalone controllers provide partial automation or remote control features, these solutions are often limited in scope and compatibility. They may monitor a single appliance or a small set of devices, without offering centralized data aggregation or comprehensive energy analysis. Many existing solutions also require manual input for scheduling and fail to provide adaptive control based on real-time electrical parameters. Furthermore, most systems do not integrate real-time alerts, mobile application connectivity, or IoT-based monitoring, leaving users with minimal visibility and control over energy usage patterns.

Another limitation of the existing system is the lack of data-driven insights. Users cannot identify high-energy-consuming devices or understand peak consumption periods effectively. This makes it challenging to implement energy-saving strategies or reduce electricity costs. The absence of automated control mechanisms and predictive alerts limits the system's reliability and practicality for modern smart homes.

In summary, the current electrical management systems in residential environments are primarily manual, limited in automation, and lack real-time monitoring, intelligent control, and IoT integration. They fail to provide users with actionable information to optimize energy consumption, ensure safety, and reduce costs. These limitations highlight the need for an integrated, intelligent solution that monitors, analyzes, and controls electricity usage efficiently while providing remote access and proactive safety alerts.

DISADVANTAGES OF EXISTING SYSTEM:

1. Manual operation of appliances leads to energy wastage.

2. No real-time monitoring of electrical parameters.
3. Limited or no automation of devices.
4. Lack of safety alerts for overcurrent or voltage fluctuations.
5. Inability to monitor individual appliance consumption.
6. Minimal integration with mobile applications or IoT.
7. No data-driven insights for energy optimization.
8. Limited scalability for additional devices.
9. Higher electricity bills due to inefficient usage.
10. Potential risk of electrical hazards remains unmitigated.

PROPOSED SYSTEM

The Smart Home Electricity Monitoring and Control System is proposed to address the limitations of existing residential electrical management systems by combining real-time monitoring, automated control, and IoT-based remote access. The system utilizes an ESP32 microcontroller, which functions as the central processing unit for gathering electrical data, controlling appliances, and communicating with a mobile application. Dedicated voltage and current sensors continuously measure electrical parameters for each connected device, providing real-time insights into energy consumption patterns. This level of monitoring allows users to track electricity usage on a granular level and identify appliances that consume excessive energy.

Appliances such as lights, fans, and other household devices are controlled using relay modules, enabling both automated operation based on pre-programmed schedules and remote control through a mobile application. The automation feature ensures that devices are only active when necessary, reducing unnecessary energy wastage. The LCD interface displays live electrical data, allowing users to observe consumption trends directly. In addition, a buzzer alert system notifies users of abnormal conditions such as overcurrent, voltage fluctuations, or short circuits, preventing potential damage to appliances and

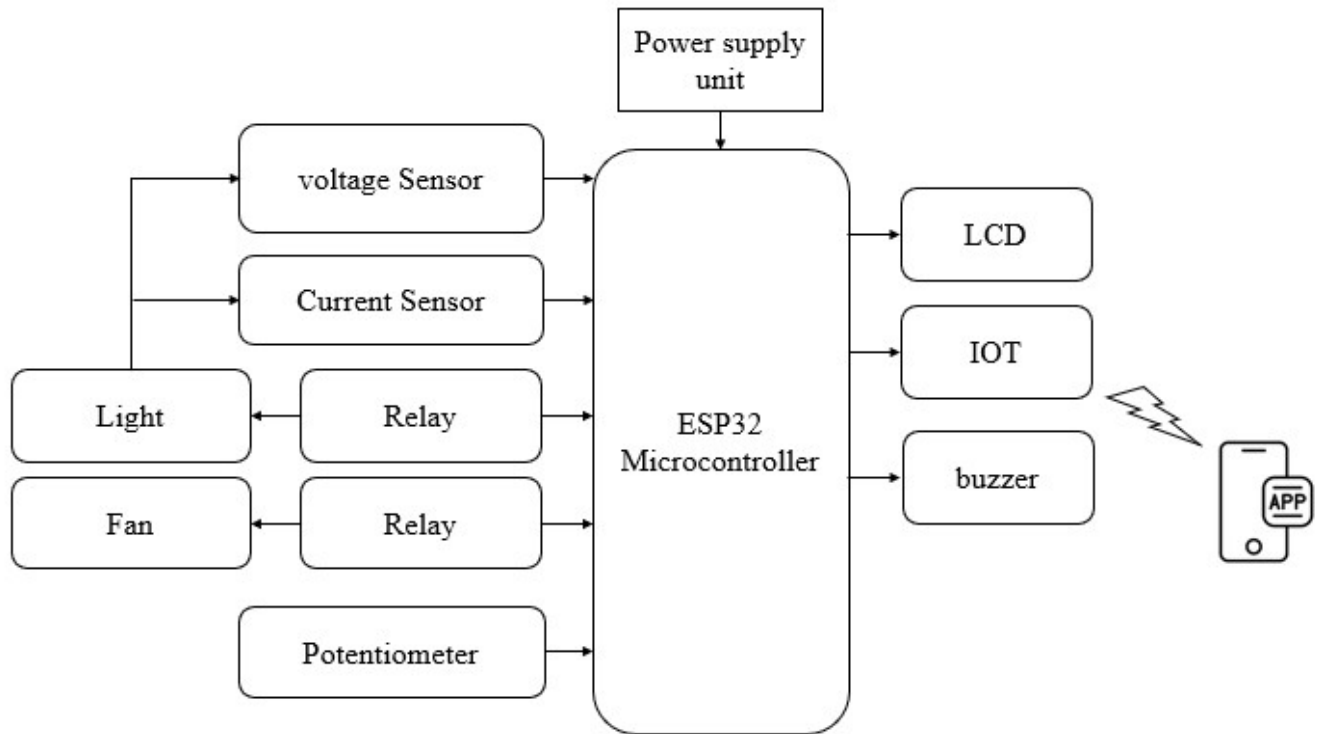
reducing safety risks.

IoT integration is a key component of the proposed system, providing users with the ability to monitor and control their home appliances from any location. Real-time data is transmitted to a mobile application, enabling users to remotely switch devices on or off, view energy usage analytics, and receive immediate alerts for abnormal conditions. The system can be further expanded to include additional smart devices, renewable energy sources, and advanced energy-saving algorithms.

The proposed system also incorporates intelligence for energy optimization. By analyzing historical data, it can suggest usage patterns, recommend appliance schedules, and detect high-energy-consuming devices. The proposed system includes a machine learning-based prediction module that analyzes energy consumption data and estimates the monthly electricity bill. This helps users understand their usage patterns and take necessary steps to reduce energy consumption. This feature supports cost reduction and promotes energy-efficient behavior. Furthermore, the system is designed to be scalable, allowing easy integration of new appliances or modules without significant reconfiguration.

Overall, the proposed system addresses the inefficiencies of existing systems by providing a centralized, automated, and remotely accessible solution that improves energy efficiency, enhances electrical safety, and increases user convenience. Its combination of real-time sensing, intelligent control, and wireless communication makes it suitable for modern smart home applications.

BLOCK DIAGRAM



ADVANTAGES OF PROPOSED SYSTEM:

1. Real-time monitoring of voltage, current, and power consumption.
2. Automated control of appliances through relay modules.
3. Remote access and control via a mobile application.
4. Alerts for abnormal electrical conditions using buzzer notifications.
5. LCD interface for direct observation of consumption data.
6. Energy optimization through usage analytics and intelligent recommendations.
7. Scalable system capable of integrating additional devices or sensors.
8. Reduction in electricity wastage and overall energy costs.
9. Enhanced electrical safety by preventing overloads and hazards.
10. User-friendly and suitable for modern smart home environments.

CHAPTER 3

3. REQUIREMENT SPECIFICATIONS

3.1. HARDWARE AND SOFTWARE SPECIFICATION

3.1.1. HARDWARE REQUIREMENTS

4. LITHIUM-ION (LI-ION) BATTERIES

Lithium-ion (Li-ion) batteries have become one of the most widely used energy storage technologies in the modern world. Their high energy density, long cycle life, and relatively low self-discharge rate make them ideal for a wide range of applications, from portable electronics to electric vehicles (EVs) and renewable energy storage systems. Since their commercialization in the early 1990s, Li-ion batteries have revolutionized the way we store and use energy, enabling advancements in technology and contributing to the global shift toward cleaner energy solutions.

This report provides a comprehensive description of lithium-ion batteries, covering their working principles, components, advantages, disadvantages, and applications. Additionally, it explores the challenges and future prospects of this technology.



1. Working Principle of Lithium-Ion Batteries

Lithium-ion batteries operate based on the movement of lithium ions between the anode and cathode during charging and discharging cycles. The fundamental principle involves electrochemical reactions that store and release energy.

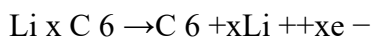
Charging Process: During charging, lithium ions move from the cathode to the anode through the electrolyte. The anode, typically made of graphite, stores the lithium ions in its layered structure. Electrons flow through

the external circuit, providing energy to the battery.

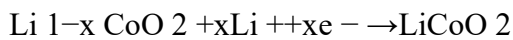
Discharging Process: During discharging, lithium ions move back from the anode to the cathode, releasing energy in the form of electrical current. The electrons flow through the external circuit, powering the connected device.

The chemical reactions at the electrodes are as follows:

Anode (Graphite):



Cathode (Lithium Cobalt Oxide, LiCoO_2):



The electrolyte, typically a lithium salt in an organic solvent, facilitates the movement of lithium ions while preventing the flow of electrons, ensuring efficient energy transfer.

2. Components of Lithium-Ion Batteries

A lithium-ion battery consists of several key components, each playing a critical role in its operation:

Anode: The anode is usually made of graphite, which has a layered structure that allows lithium ions to intercalate (embed) during charging. The anode stores lithium ions and releases them during discharging.

Cathode: The cathode is typically made of lithium metal oxides, such as lithium cobalt oxide (LiCoO_2), lithium iron phosphate (LiFePO_4), or lithium nickel manganese cobalt oxide (NMC). The cathode material determines the battery's voltage, capacity, and thermal stability.

Electrolyte: The electrolyte is a lithium salt dissolved in an organic solvent. It acts as a medium for lithium ion transport between the anode and cathode. The electrolyte must be chemically stable and have high ionic conductivity.

Separator: The separator is a porous membrane that prevents physical contact between the anode and cathode while allowing lithium ions to pass through. It is typically made of polyethylene or polypropylene.

Current Collectors: These are conductive materials (usually aluminum for the cathode and copper for the anode) that collect and distribute the electrical current generated during the electrochemical reactions.

Casing: The casing encloses the battery components and provides mechanical support and protection. It is usually made of metal or polymer materials.

3. Advantages of Lithium-Ion Batteries

Lithium-ion batteries offer several advantages over other types of rechargeable batteries, making them the preferred choice for many applications:

High Energy Density: Li-ion batteries have a high energy density, meaning they can store a large amount of energy in a relatively small and lightweight package. This makes them ideal for portable electronics and electric vehicles.

Long Cycle Life: Li-ion batteries can endure hundreds to thousands of charge-discharge cycles before their capacity significantly degrades. This long cycle life reduces the need for frequent replacements.

Low Self-Discharge Rate: Compared to other rechargeable batteries, Li-ion batteries have a low self-discharge rate, meaning they retain their charge for longer periods when not in use.

Fast Charging: Li-ion batteries can be charged at a relatively fast rate, making them convenient for applications where quick charging is essential.

No Memory Effect: Unlike some other rechargeable batteries, Li-ion batteries do not suffer from the memory effect, which can reduce the effective capacity of the battery over time.

Environmentally Friendly: Li-ion batteries do not contain toxic heavy metals like lead or cadmium, making them more environmentally friendly compared to some other battery types.

4. Disadvantages and Challenges of Lithium-Ion Batteries

Despite their many advantages, lithium-ion batteries also have some limitations and challenges:

Safety Concerns: Li-ion batteries are susceptible to thermal runaway, a condition where excessive heat causes the battery to catch fire or explode. This is often caused by overcharging, physical damage, or manufacturing defects.

Cost: The materials used in Li-ion batteries, such as cobalt and lithium, are expensive. This makes the batteries costly to produce, although prices have been decreasing over time.

Limited Lifespan: While Li-ion batteries have a long cycle life, they eventually degrade over time, especially when exposed to high temperatures or frequent deep discharges.

Resource Availability: The production of Li-ion batteries relies on finite resources like lithium and cobalt. The mining of these materials can have environmental and ethical implications, such as habitat destruction and labor issues.

Recycling Challenges: Recycling Li-ion batteries is complex and expensive due to the variety of materials used and the need to safely handle hazardous components. However, advancements in recycling technologies are being made to address this issue.

5. Applications of Lithium-Ion Batteries

Lithium-ion batteries are used in a wide range of applications due to their high energy density, long cycle life, and versatility. Some of the key applications include:

Consumer Electronics: Li-ion batteries are widely used in smartphones, laptops, tablets, and other portable electronic devices due to their compact size and high energy density.

Electric Vehicles (EVs): The automotive industry has embraced Li-ion batteries for electric and hybrid vehicles because of their ability to provide high power and long driving ranges.

Renewable Energy Storage: Li-ion batteries are used to store energy generated from renewable sources like

solar and wind, enabling a stable power supply even when the sun isn't shining or the wind isn't blowing.

Medical Devices: Portable medical devices, such as insulin pumps and defibrillators, rely on Li-ion batteries for their reliability and long life.

Aerospace and Defense: Li-ion batteries are used in satellites, drones, and military equipment due to their lightweight and high energy output.

POWER SUPPLY

The 7812 and 7805 voltage regulators are commonly used components to provide stable DC voltage outputs of +12V and +5V, respectively, from a higher input voltage source.



7812 Voltage Regulator:

Input Voltage: Typically requires an input voltage slightly higher than 12V (usually around 14-16V) to regulate effectively.

Output Voltage: Provides a stable +12V DC output.

Capacitors:

C1000/25: This likely refers to a capacitor with a capacitance of 1000 μ F and a voltage rating of 25V. This capacitor is typically placed on the input side (between input and ground) to stabilize the input voltage, reducing noise and providing a reservoir of charge to handle transient spikes.

C10/63: This could refer to a capacitor with a capacitance of 10 μ F and a voltage rating of 63V. This capacitor is usually placed on the output side (between output and ground) to stabilize the output voltage, filtering out any remaining noise and improving regulation.

Resistor: A resistor isn't typically used directly with the 7812 regulator in the same way as capacitors are, but it can be part of the circuit design for specific applications, such as in voltage dividers or as part of a

feedback loop for stability.

7805 Voltage Regulator:

Input Voltage: Requires an input voltage typically around 7-25V (ideal 7-20V) to regulate effectively.

Output Voltage: Provides a stable +5V DC output.

Capacitors:

C1000/25: As with the 7812, this capacitor stabilizes the input voltage to the regulator.

C10/63: This capacitor stabilizes the output voltage of the regulator.

Resistor: Similar to the 7812, resistors are not directly part of the typical configuration but can be used in specific applications.

Circuit Considerations:

Decoupling Capacitors: These capacitors (like C1000/25 and C10/63) are crucial for filtering out noise and stabilizing the voltage levels, ensuring reliable operation of your circuit.

Heat Dissipation: Both regulators can generate heat, especially when dropping significant voltage. Adequate heat sinking may be necessary depending on the current drawn and the input-output voltage differential.

Current Requirements: Ensure that the regulators can supply enough current for your application. If higher currents are required, additional heat sinking and possibly parallel regulators may be needed.

In summary, the 7812 and 7805 voltage regulators, along with capacitors like C1000/25 and C10/63, form a basic yet effective setup for providing stable +12V and +5V outputs in electronic circuits, suitable for a wide range of applications from powering microcontrollers to analog circuits.

ESP32 MICROCONTROLLER

The ESP32 is an exceptionally versatile and powerful microcontroller designed for a wide range of Internet of Things (IoT) applications. Developed by Espressif Systems, it integrates advanced features, including built-in Wi-Fi and Bluetooth capabilities, which facilitate seamless connectivity to the internet and other devices. At its core, the ESP32 is powered by a dual-core Tensilica LX6 processor, capable of running at speeds up to 240 MHz, enabling it to handle multiple tasks simultaneously. This makes it suitable for complex applications that require significant processing power, such as real-time data analytics and control systems.

With 520 KB of SRAM and the ability to support external SPI flash memory of up to 16 MB, the ESP32 can accommodate extensive programming and data logging requirements. Its rich set of peripherals includes up to 34 programmable GPIO pins, ADC (Analog-to-Digital Converters) with 12-bit resolution, and DAC (Digital-to-Analog Converters) for generating audio signals or other analog outputs. The presence of various

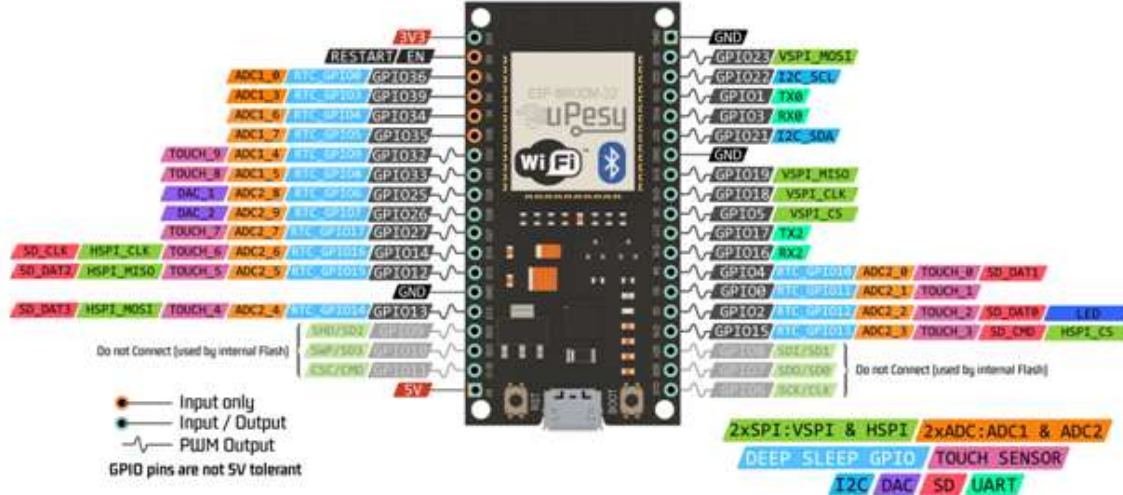
communication protocols, such as SPI, I2C, and UART, allows for easy integration with a multitude of sensors and devices, enhancing its applicability across various projects.

Power management is another strength of the ESP32, featuring several power-saving modes that are particularly beneficial for battery-operated devices. These modes allow the microcontroller to enter deep sleep states, significantly reducing power consumption when the device is idle. Furthermore, robust security features, including hardware encryption (AES, SHA-2), secure boot, and flash encryption, ensure secure communication and data integrity, which are essential for many IoT applications where data security is paramount.

The ESP32 is widely supported by various development environments, including the Arduino IDE and Espressif's own IoT Development Framework (ESP-IDF), which provide extensive libraries and examples to accelerate the development process. Its applications span diverse fields, from smart home devices that automate lighting and security systems to wearable technology for fitness tracking and health monitoring. The microcontroller is also used in environmental monitoring systems that measure temperature, humidity, and air quality, as well as in industrial automation for monitoring equipment performance and controlling automated processes. In the realm of robotics, the ESP32 serves as a central control unit, facilitating wireless communication and sensor integration for complex robotic systems.

In conclusion, the ESP32 stands out as a highly capable and adaptable microcontroller, combining performance, versatility, and extensive connectivity features in a compact package. Its dual-core architecture and comprehensive set of peripherals make it an excellent choice for a wide variety of applications in the ever-expanding field of IoT. Whether for simple hobbyist projects or sophisticated industrial solutions, the ESP32 continues to gain popularity among developers and engineers for its performance, ease of use, and robust feature set, ultimately paving the way for innovative technological advancements in numerous domains.

ESP32 DevKit Full Pinout



ESP32 PIN FUNCTIONS

- **GPIO Pins:** Up to 34 configurable General Purpose Input/Output pins for versatile applications. These pins can be programmed for digital input or output, making them suitable for interfacing with a variety of sensors and actuators.
- **ADC:** Up to 18 channels with 12-bit resolution for accurate analog signal measurement. This feature allows the ESP32 to read varying voltage levels from sensors, providing precise data for applications like environmental monitoring.
- **DAC:** 2 channels with 8-bit resolution for generating analog outputs. The DAC can be used for audio applications or to control devices that require variable voltage levels.
- **Power Pins:** V5 for 5V input, V3 for 3.3V output, and multiple GND pins. These power pins facilitate connections to external devices and ensure that the ESP32 operates within its required voltage range.
- **I2C Interface:** SDA and SCL pins for connecting I2C devices. The I2C protocol allows multiple devices to share the same bus, simplifying the wiring and connections needed for complex projects.
- **SPI Interface:** Standard pins for MISO, MOSI, SCK, and CS for SPI communication. SPI is a high-speed interface ideal for connecting fast peripherals, such as sensors and memory cards.
- **UART Interface:** TX and RX pins for serial communication. These pins enable the ESP32 to communicate with other devices over a serial interface, useful for debugging and data transmission.
- **Touch Sensor Pins:** Configurable pins for touch-sensitive input. This feature allows for the development of user interfaces that can detect touch input, enhancing interactivity in applications.

- **RTC Functions:** Pins for real-time clock operations and wake-up triggers. These functions are essential for applications requiring time-based operations, even during low power modes.
- **Programming/Debugging:** EN pin for chip enable and IO0 for boot mode selection during programming. These pins are crucial for loading new firmware and debugging the ESP32 during development.

ESP32 Specifications

- **Microprocessor:** Dual-core Tensilica LX6 processor with up to 240 MHz clock speed. This architecture allows for efficient handling of multiple tasks simultaneously, making it suitable for demanding applications.
- **Memory:** 520 KB of SRAM and support for up to 16 MB of external SPI flash memory. The ample memory capacity enables the execution of complex algorithms and the storage of extensive data.
- **Wireless Connectivity:** Integrated Wi-Fi (IEEE 802.11 b/g/n) and Bluetooth (Classic and BLE). This dual connectivity enhances the flexibility of the ESP32 in IoT applications, allowing seamless communication with various devices.
- **GPIO Pins:** Up to 34 General Purpose Input/Output pins for various applications. These pins can serve multiple functions, such as analog input, PWM output, or digital signaling, making the ESP32 adaptable to different project needs.
- **Analog Features:** 12-bit ADC with up to 18 channels and 2 channels of 8-bit DAC. The high-resolution analog features are ideal for applications requiring precise signal measurement and generation.
- **Power Supply:** Operates within a voltage range of 3.0V to 3.6V. This range allows the ESP32 to be powered by various sources, including batteries and regulated power supplies, enhancing its usability in portable applications.
- **Power Management:** Multiple power-saving modes, including deep sleep. These modes significantly reduce power consumption, making the ESP32 suitable for battery-operated devices and energy-efficient applications.
- **Operating Temperature:** Functional range from -40°C to 85°C. This wide temperature range ensures reliable performance in diverse environmental conditions, from industrial applications to outdoor deployments.
- **Security Features:** Hardware encryption (AES, SHA-2), secure boot, and flash encryption. These security measures help protect sensitive data and ensure the integrity of the device, which is critical in IoT applications.

- **Development Support:** Compatible with Arduino IDE, Espressif ESP-IDF, and other frameworks. The extensive development support, along with a rich set of libraries and resources, simplifies the programming process and accelerates project development.

LIQUID CRYSTAL DISPLAY

A liquid crystal display (LCD) is a flat panel display, electronic visual display, or video display that uses the light modulating properties of liquid crystals. Liquid crystals do not emit light directly. LCDs are available to display arbitrary images (as in a general-purpose computer display) or fixed images which can be displayed or hidden, such as preset words, digits, and 7-segment displays as in a digital clock. They use the same basic technology, except that arbitrary images are made up of a large number of small pixels, while other displays have larger elements. An LCD is a small low cost display. It is easy to interface with a micro-controller because of an embedded controller (the black blob on the back of the board). This controller is standard across many displays (HD 44780) which means many micro-controllers (including the Arduino) have libraries that make displaying messages as easy as a single line of code.



LCD display unit

LCDs are used in a wide range of applications including computer monitors, televisions, instrument panels, aircraft cockpit displays, and signage. They are common in consumer devices such as video players, gaming devices, clocks, watches, calculators, and telephones, and have replaced cathode ray tube (CRT) displays in most applications. They are available in a wider range of screen sizes than CRT and plasma displays, and since they do not use phosphors, they do not suffer image burn-in. LCDs are, however, susceptible to image persistence.

16X2 LCD SPECIFICATIONS

- **Display Format:** 16 characters per line, 2 lines total.

- Character Size: 5x8 pixels for standard characters.
- Dimensions: Approximately 80mm x 36mm x 13mm.
- Interface: Parallel (4-bit or 8-bit mode).
- Supply Voltage: Typically 5V DC.
- Current Consumption: Around 1.5 mA at 5V.
- Backlight: LED backlight (3.3V to 5V).
- Temperature Range: 0°C to 70°C operating, -20°C to 80°C storage.
- Response Time: Under 10 ms.
- Mounting: PCB or breadboard compatible.
- Character Set: Standard ASCII with custom character support.

I2C (Inter-Integrated Circuit)

The I2C connection for an LCD display enables efficient and simplified interfacing with microcontrollers, like the ESP32, by requiring only two data lines: SDA (data) and SCL (clock). This setup significantly reduces the number of pins needed, compared to traditional parallel connections, allowing other I/O pins on the microcontroller to remain free for additional sensors or peripherals. The I2C interface is controlled by a small I2C module on the back of the LCD, typically using a chip like the PCF8574, which converts the I2C signals into parallel signals the LCD can interpret.

Each device on the I2C bus has a unique address (e.g., 0x27 for many LCD modules), letting the microcontroller communicate with specific devices even if there are multiple I2C components on the same bus. The power (VCC) and ground (GND) lines complete the connections, and the data transfer is synchronous, meaning the microcontroller (master) sends commands in sync with the LCD (slave). Commands like positioning the cursor, clearing the display, or updating text are all handled through simple I2C libraries that streamline programming. This makes I2C ideal for projects that require organized wiring and efficient communication, especially in setups where pin limitations and space are considerations.

VOLTAGE SENSOR:

A voltage sensor is a device used to measure and monitor the electrical potential difference (voltage) in a circuit. These sensors are crucial in various applications, including power systems, industrial automation, and electronic devices, as they help ensure the proper functioning and safety of electrical systems.



Types of Voltage Sensors:

- **Analog Voltage Sensors:** Provide a continuous output signal proportional to the measured voltage. They are commonly used in simple applications.
- **Digital Voltage Sensors:** Convert the measured voltage into a digital signal for precise measurement and easy interfacing with digital systems.
- **AC Voltage Sensors:** Specifically designed to measure alternating current (AC) voltage.
- **DC Voltage Sensors:** Specifically designed to measure direct current (DC) voltage.

Working Principle:

Voltage sensors work based on various principles, including resistive dividers, capacitive dividers, and inductive coupling. The most common method involves a resistive divider, where the sensor reduces the voltage to a measurable level, which can then be processed by a microcontroller or other monitoring device.

Voltage Sensor Pin Configuration:

- **VCC (Power Supply Pin):** Connect this pin to the positive voltage supply. For most sensors, this is typically 5V or 3.3V, depending on your sensor's requirements.
- **GND (Ground Pin):** Connect this pin to the ground (GND) of your microcontroller or development board.
- **VOOUT (Output Voltage Pin):** This pin provides the output voltage corresponding to the measured input voltage. Connect this pin to an analog input pin on your microcontroller to read the voltage value.

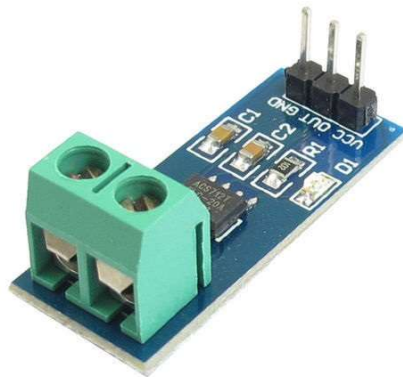
Applications:

- **Power Monitoring:** To monitor voltage levels in power systems and ensure stable power supply.
- **Battery Management:** To measure and monitor battery voltage in electric vehicles and portable electronics.
- **Industrial Automation:** To monitor voltage levels in machinery and control systems.

- **Renewable Energy Systems:** To monitor and control voltage levels in solar and wind power systems.

CURRENT SENSOR:

A current sensor is a device used to detect and measure the amount of current flowing through a conductor. These sensors are crucial in various applications, such as power monitoring, battery management, motor control, and energy metering.



Types of Current Sensors

- **Shunt Resistor Sensors:** Use a precision resistor to measure voltage drop, which is proportional to the current flow.
- **Hall Effect Sensors:** Utilize the Hall effect principle to measure the magnetic field generated by current flow.
- **Current Transformer (CT) Sensors:** Inductive devices used to measure alternating current (AC) by transforming high currents to a lower, more manageable level.
- **Rogowski Coil Sensors:** Flexible coils used to measure AC currents without physical contact with the conductor.
- **Magnetoresistive Sensors:** Measure changes in resistance due to the magnetic field generated by current flow.

Working Principle

Current sensors work by detecting the magnetic field generated by the current or by measuring the voltage drop across a known resistor. The output is typically an analog signal proportional to the current, which can be processed by a microcontroller or other monitoring device.

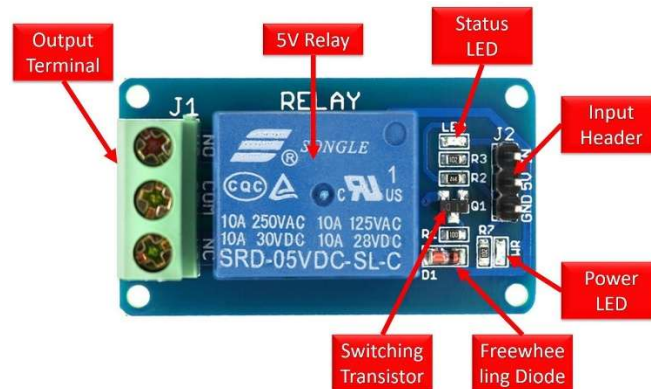
Applications

- **Power Monitoring:** To measure current consumption in power systems and ensure efficient energy usage.
- **Battery Management:** To monitor charging and discharging currents in battery-powered devices.

- **Motor Control:** To measure and control the current supplied to motors in industrial and automotive applications.
- **Energy Metering:** To measure current consumption in residential, commercial, and industrial settings.
- **Renewable Energy Systems:** To monitor current flows in solar panels and wind turbines.

RELAY:

A relay is an electromechanical device used to control high-power electrical devices with low-power signals, such as those from a microcontroller like Arduino. It works by using a small current to control a larger current, enabling you to switch circuits on and off safely.



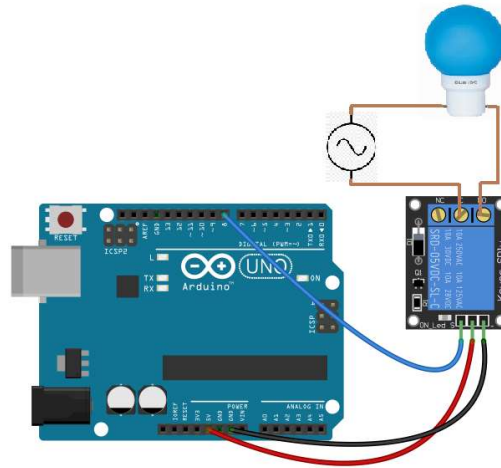
Wiring:

Here's a general wiring setup using a relay module with an Arduino:

- **VCC and GND:** Connect the VCC and GND pins of the relay module to the 5V and GND pins on the Arduino, respectively.
- **Control Signal:** Connect the control pin (often labeled as IN or SIG) of the relay module to a digital output pin on the Arduino (e.g., pin 7).

Load and Relay Contacts:

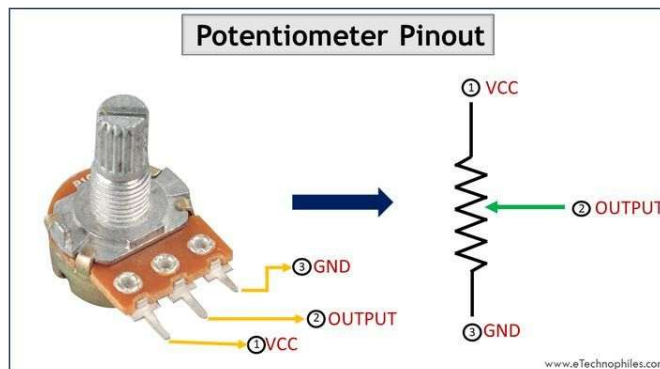
- Connect one terminal of your high-power device (load) to the power supply (usually mains voltage).
- Connect the other terminal of the load to the Common (COM) terminal of the relay.



- Connect the Normally Open (NO) terminal of the relay to the other side of the power supply.
- If needed, connect the Normally Closed (NC) terminal to the other side of the load for alternative switching configurations.

POTENTIOMETER:

A potentiometer, often referred to simply as a "pot," is a three-terminal variable resistor that allows for adjustable electrical resistance. It's commonly used in electronics to control various parameters such as volume, brightness, speed, and other analog settings.



Structure and Operation

Three Terminals:

- Potentiometers typically have three terminals: two outer terminals and one wiper terminal (the movable contact).
- The resistance between the two outer terminals is fixed and determines the total resistance of the potentiometer.
- The wiper terminal is adjustable and moves along a resistive track between the two outer terminals.

Variable Resistance:

- By rotating the shaft or knob of the potentiometer, the position of the wiper changes along the resistive track.
- This movement changes the resistance between the wiper terminal and either of the outer terminals, creating a variable resistance.

The pin configuration of a typical potentiometer depends on its type and design, but here's a general outline of the standard pin connections for a potentiometer:

Outer Terminals:

There are two outer terminals, often labeled as "1" and "3".

Terminal 1 (often marked as "CCW" or "CW") is typically connected to the outer end of the resistive track.

Terminal 3 (often marked as "CW" or "CCW") is connected to the other end of the resistive track.

Wiper Terminal:

The middle terminal, labeled as "2" or sometimes with an arrow symbol, is the wiper terminal.

The wiper moves along the resistive track when the potentiometer knob or shaft is rotated.

Applications:

- Voltage Divider: Used in voltage divider circuits to generate a variable output voltage based on the position of the wiper.
- Control Input: Provides an adjustable input signal to control parameters in electronic circuits (e.g., volume control in audio amplifiers).
- Sensor Adjustment: Adjusts sensor readings or analog signals in instrumentation and control systems.

SOFTWARE REQUIREMENTS

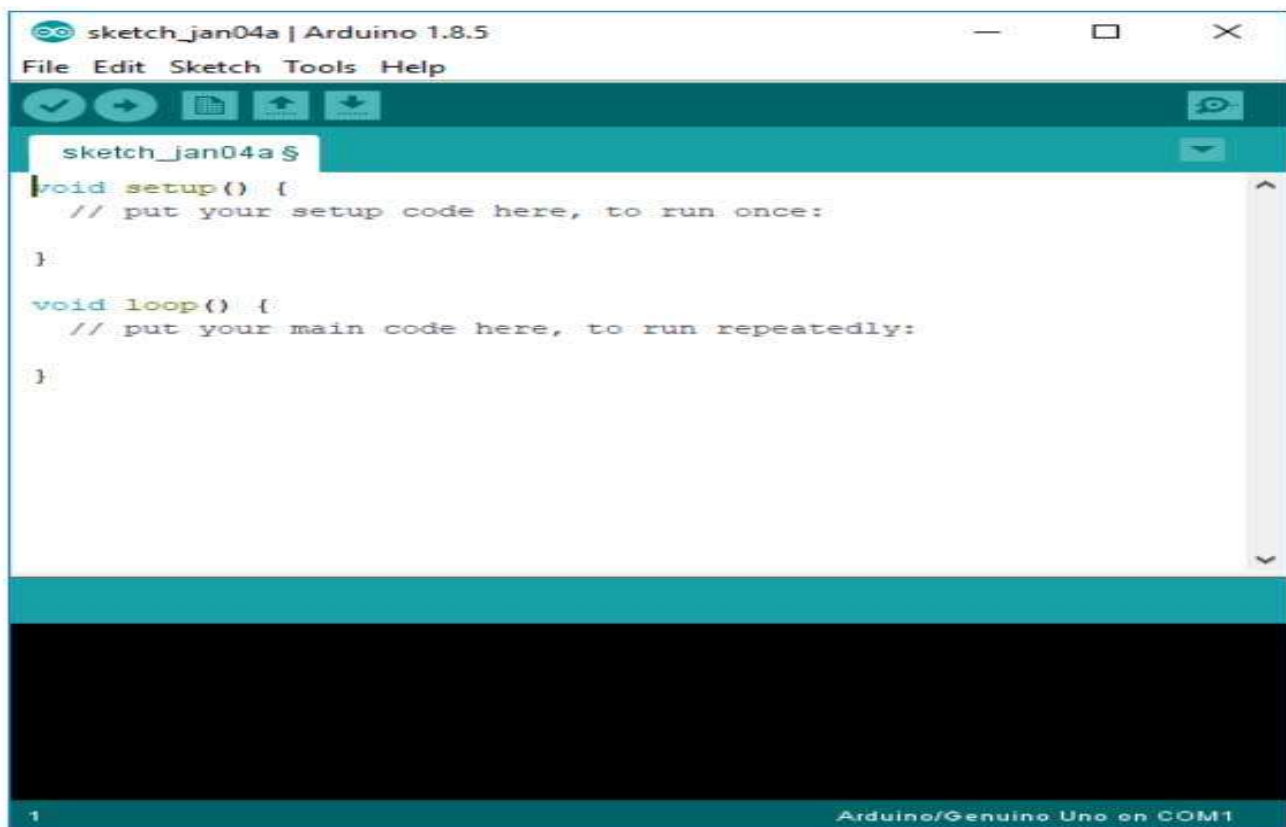
ARDUINO IDE

The Arduino integrated development environment (IDE) is a cross-platform application (for Windows, macOS, Linux) that is written in the programming language Java. It is used to write and upload programs to Arduino board.

The source code for the IDE is released under the GNU General Public License, version 2. The Arduino IDE supports the languages [C](#) and C++ using special rules of code structuring.^[4] The Arduino IDE supplies a software library from the Wiring project, which provides many common input and output

procedures. User-written code only requires two basic functions, for starting the sketch and the main program loop, that are compiled and linked with a program stub `main()` into an executable cyclic executive program with the GNU tool chain, also included with the IDE distribution. The Arduino IDE employs the program argued to convert the executable code into a text file in hexadecimal encoding that is loaded into the Arduino board by a loader program in the board's firmware.

Arduino is an open-source electronics platform based on easy-to-use hardware and software. Arduino boards are able to read inputs - light on a sensor, a finger on a button, or a Twitter message - and turn it into an output - activating a motor, turning on an LED, publishing something online. You can tell your board what to do by sending a set of instructions to the microcontroller on the board



The Arduino IDE

The Arduino IDE is incredibly minimalistic, yet it provides a near-complete environment for most Arduino-based projects. The top menu bar has the standard options, including "File" (new, load save, etc.), "Edit" (font, copy, paste, etc.), "Sketch" (for compiling and programming), "Tools" (useful options for testing projects), and "Help". The middle section of the IDE is a simple text editor that where you can enter the program code. The bottom section of the IDE is dedicated to an output window that is used to see the status of the compilation, how much memory has been used, any errors that were found in the

program, and various other useful messages.

Projects made using the Arduino are called sketches, and such sketches are usually written in a cut-down version of C++ (a number of C++ features are not included). Because programming a microcontroller is somewhat different from programming a computer, there are a number of device-specific libraries (e.g., changing pin modes, output data on pins, reading analog values, and timers). This sometimes confuses users who think Arduino is programmed in an “Arduino language.” However, the Arduino is, in fact, programmed in C++. It just uses unique libraries for the device.

The Arduino Integrated Development Environment - or Arduino Software (IDE) - contains a text editor for writing code, a message area, a text console, a toolbar with buttons for common functions and a series of menus. It connects to the Arduino and Genuine hardware to upload programs and communicate with them.

Programs written using Arduino Software (IDE) are called **sketches**. These sketches are written in the text editor and are saved with the file extension `.ino`. The editor has features for cutting/pasting and for searching/replacing text. The message area gives feedback while saving and exporting and also displays errors. The console displays text output by the Arduino Software (IDE), including complete error messages and other information. The bottom right hand corner of the window displays the configured board and serial port. The toolbar buttons allow you to verify and upload programs, create, open, and save sketches, and open the serial monitor.

LIBRARIES

Libraries provide extra functionality for use in sketches, e.g. working with hardware or manipulating data. To use a library in a sketch, select it from the **Sketch > Import Library** menu. This will insert one or more **#include** statements at the top of the sketch and compile the library with your sketch. Because libraries are uploaded to the board with your sketch, they increase the amount of space it takes up. If a sketch no longer needs a library, simply delete its **#include** statements from the top of your code.

There is a list of libraries in the reference. Some libraries are included with the Arduino software. Others can be downloaded from a variety of sources or through the Library Manager. Starting with version 1.0.5 of the IDE, you do can import a library from a zip file and use it in an open sketch.

CONNECTING THE ARDUINO

Connecting an Arduino board to your PC is quite simple. On Windows:

1. Plug in the USB cable - one end to the PC, and one end to the Arduino board.

2. When prompted, select "Browse my computer for driver" and then select the folder to which you extracted your original Arduino IDE download.
3. You may receive an error that the board is not a Microsoft certified device - select "Install anyway."
4. Your board should now be ready for programming.

When programming your Arduino board it is important to know what COM port the Arduino is using on your PC. On Windows, navigate to Start->Devices and Printers, and look for the Arduino. The COM port will be displayed underneath.

Alternatively, the message telling you that the Arduino has been connected successfully in the lower-left hand corner of your screen usually specifies the COM port it is using.

PREPARING THE BOARD

Before loading any code to your Arduino board, you must first open the IDE. Double click the Arduino .exe file that you downloaded earlier. A blank program, or "sketch," should open.

The Blink example is the easiest way to test any Arduino board. Within the Arduino window, it can be found under File->Examples->Basics->Blink.

Before the code can be uploaded to your board, two important steps are required.

1. Select your Arduino from the list under Tools->Board. The standard board used in RBE 1001, 2001, and 2002 is the Arduino Mega 2560, so select the "Arduino Mega 2560 or Mega ADK" option in the dropdown.
2. Select the communication port, or COM port, by going to Tools->Serial Port.

If you noted the COM port your Arduino board is using, it should be listed in the dropdown menu. If not, your board has not finished installing or needs to be reconnected.

LOADING CODE

The upper left of the Arduino window has two buttons: A checkmark to Verify your code, and a right-facing arrow to Upload it. Press the right arrow button to compile and upload the Blink example to your Arduino board.

The black bar at the bottom of the Arduino window is reserved for messages indicating the success or failure of code uploading. A "Completed Successfully" message should appear once the code is done

uploading to your board. If an error message appears instead, check that you selected the correct board and COM port in the Tools menu, and check your physical connections.

If uploaded successfully, the LED on your board should blink on/off once every second. Most Arduino boards have an LED prewired to pin 13.

It is very important that you do not use pins 0 or 1 while loading code. It is recommended that you do not use those pins ever.

Arduino code is loaded over a serial port to the controller. Older models use an [FTDI](#) chip which deals with all the USB specifics. Newer models have either a small AVR that mimics the FTDI chip or a built-in USB-to-serial port on the AVR micro-controller itself.

5.2 PROTEUS



Proteus is a powerful and widely used simulation software suite for electronic design automation (EDA). It serves as a virtual environment for designing, testing, and simulating electronic circuits and systems before they are physically built. Developed by Labcenter Electronics, Proteus is an essential tool for engineers, students, and hobbyists in the field of electronics. One of Proteus strengths lies in its comprehensive component library. It provides a vast collection of electronic components, including microcontrollers, sensors, actuators, analog and digital integrated circuits, and various passive components. This extensive library allows users to create complex and diverse electronic circuits, ranging from simple LED blinkers to advanced microcontroller-based systems, with ease. Additionally, Proteus allows users to create custom components, providing flexibility for unique and specialized projects. Its ability to simulate the behavior of electronic circuits accurately. It employs advanced algorithms to model the interactions between components, taking into account parameters such as resistance, capacitance, inductance, and semiconductor characteristics. This enables engineers to analyze circuit performance, troubleshoot potential issues, and optimize designs without the need for physical prototyping. Moreover, Proteus supports both analog and digital simulation, allowing for a comprehensive evaluation of mixed-signal systems. The feature of Proteus is its support for microcontroller-based designs. It includes a wide range of virtual microcontrollers, covering popular families like the PIC, AVR, Arduino, and more. Users can write and upload firmware code to these virtual

microcontrollers, enabling them to interact with the simulated circuit in a manner similar to real-world applications. This capability is invaluable for testing and validating embedded systems before they are implemented in hardware. Proteus also offers a powerful debugging and visualization environment. It provides tools like oscilloscopes, logic analyzers, and virtual instrumentation panels, which allow users to observe and analyze the behavior of their circuits in real-time during simulation. This feature is crucial for identifying and rectifying issues that may arise during the design process. Proteus supports hardware-in-the-loop (HIL) testing, enabling users to interface their virtual circuits with real-world hardware.

CHAPTER 4

4.METHODOLOGY

The methodology of the Smart Home Electricity Monitoring and Control System focuses on the systematic approach for designing, implementing, and testing a reliable and efficient solution for monitoring and controlling household electricity consumption. The methodology integrates sensor-based data acquisition, real-time processing with the ESP32 microcontroller, appliance automation using relays, user interaction via an LCD display, and IoT-based remote monitoring. The workflow is divided into distinct phases, including system design, hardware implementation, software development, IoT integration, and testing.

1. System Design and Architecture

The initial phase involves the conceptualization and design of the system architecture. The architecture is divided into four key components: sensing, control, display, and communication. Sensing involves measuring electrical parameters such as voltage, current, and power consumption using precise sensors. These sensors provide real-time feedback to the ESP32 microcontroller, which acts as the central processing unit. The control unit consists of relay modules that regulate the operation of electrical appliances. The display unit employs an LCD to provide users with real-time data on energy consumption and appliance status. Finally, the communication unit leverages IoT technology to transmit real-time data to a mobile application, allowing users to monitor and control their appliances remotely.

2. Hardware Implementation

Hardware selection and integration are critical to ensure accuracy, reliability, and scalability. The ESP32 microcontroller was chosen for its high processing speed, built-in Wi-Fi capability, and compatibility with multiple sensor modules. Voltage and current sensors are interfaced with the microcontroller to measure electrical parameters in real

time. Relay modules are connected to appliances such as lights and fans to enable automated and remote switching. An LCD is installed to display the live voltage, current, and power consumption data. A buzzer is also connected to alert users in case of abnormal conditions, such as overcurrent, voltage spikes, or short circuits. All components are integrated into a centralized circuit layout to ensure compactness and reduce wiring complexity.

3. Software Development

The software development phase involves programming the ESP32 microcontroller to process sensor inputs, control relays, and communicate with the IoT platform. The microcontroller continuously reads data from the voltage and current sensors and calculates instantaneous power consumption. Threshold limits are defined to detect abnormal conditions, triggering buzzer alerts when necessary. The software also allows automated control of appliances based on predefined schedules or sensor readings. For IoT integration, the system uses MQTT or HTTP protocols to transmit data to the mobile application in real time. A user-friendly interface is implemented on the application, allowing remote switching of appliances, viewing of live energy data, and receiving alerts.

4. IoT Integration and Remote Monitoring

IoT integration is a major feature that enhances system functionality. Real-time data collected from sensors is transmitted via the ESP32's Wi-Fi module to a cloud server or IoT platform. Users can access the mobile application to monitor electricity usage, control appliances, and receive instant notifications. The system ensures secure and reliable communication by using authentication protocols. Historical data is stored for analysis, allowing users to track energy consumption trends over time and make informed decisions to reduce power wastage. The IoT framework also enables scalability, allowing the addition of more appliances or sensors without significant redesign.

5. Data Analysis and Energy Optimization

An important aspect of the methodology is analyzing the collected energy data to optimize power usage. The system records consumption data for each connected appliance and

identifies high-energy-consuming devices. Algorithms are employed to suggest optimal operation schedules, such as turning off appliances during peak hours or switching devices to energy-saving modes. This data-driven approach reduces electricity bills, promotes sustainable energy usage, and encourages users to adopt energy-efficient habits.

6. Safety Mechanisms

The proposed system incorporates several safety mechanisms to prevent electrical hazards. Overcurrent and voltage spike detection are implemented using sensor readings and pre-defined thresholds. When abnormal conditions are detected, the system triggers a buzzer alert and sends notifications to the user via the mobile application. Relay modules can also automatically disconnect appliances during critical events, preventing damage to devices and reducing fire risks. By integrating these mechanisms, the system ensures both safety and reliability.

7. Testing and Validation

Once hardware and software are integrated, comprehensive testing is conducted to validate system performance. The testing includes monitoring the accuracy of voltage, current, and power measurements, verifying relay operation for appliance control, and checking the responsiveness of buzzer alerts. IoT communication is tested for real-time data transmission and remote appliance control. Stress tests are performed under varying electrical loads to ensure system stability and reliability. Feedback from test runs is used to optimize software algorithms and calibrate sensors for accurate measurements.

8. Scalability and Future Integration

The methodology also emphasizes scalability. The system is designed to accommodate additional appliances, smart meters, or renewable energy sources such as solar panels. Future integration may include AI-based energy prediction, machine learning for anomaly detection, and voice control through smart assistants. This forward-compatible design ensures that the system can evolve with future smart home technologies.

9. Summary of Methodology

The methodology combines sensor-based monitoring, automated control, real-time data processing, IoT connectivity, and safety mechanisms in a structured workflow. By integrating these components, the system provides users with a centralized, intelligent, and remotely accessible solution for managing household electricity. The methodology ensures energy efficiency, reduces costs, enhances safety, and improves convenience, making it suitable for modern smart home environments.

CHAPTER 5

ARCHITECTURE DIAGRAM

5.1 SYSTEM ARCHITECTURE

The system architecture of the Smart Home Electricity Monitoring and Control System is designed to provide an integrated, intelligent, and scalable framework for monitoring, controlling, and optimizing residential electricity consumption. The architecture is structured into four primary layers: the sensing layer, the control layer, the display and alert layer, and the communication layer. Each layer plays a distinct role in ensuring the efficiency, safety, and user-friendliness of the system, while collectively forming a cohesive framework for modern smart home energy management.

1. Sensing Layer

The sensing layer forms the foundation of the system architecture by providing accurate, real-time measurements of electrical parameters. This layer employs voltage and current sensors to monitor the electrical supply to individual appliances or circuits. The sensors capture instantaneous readings of voltage, current, and power consumption, which are essential for both energy monitoring and anomaly detection. The collected data is then transmitted to the ESP32 microcontroller, which functions as the central processing unit of the system. This layer ensures that all subsequent layers have reliable and precise information to process and act upon, making real-time monitoring and intelligent control feasible.

2. Control Layer

The control layer is responsible for regulating household appliances based on sensor data, pre-programmed schedules, and user commands. Relay modules are employed to switch electrical appliances such as lights, fans, and other devices on or off. Automated control is achieved through software algorithms running on the ESP32, which determine appliance operation based on energy usage patterns, threshold limits, and user preferences. In addition to automation, this layer supports manual remote control via a mobile application, allowing users to manage appliances from any location. This layer not only enhances convenience but also optimizes energy consumption by ensuring appliances operate efficiently.

3. Display and Alert Layer

The display and alert layer provides visual and auditory feedback to users about the system's status. A Liquid Crystal Display (LCD) presents real-time electrical data, including voltage, current, power consumption, and appliance status. This immediate visibility allows users to monitor household energy usage effectively. The system also integrates a buzzer for alerts, which is triggered under abnormal conditions such as overcurrent, voltage fluctuations, or short circuits. These safety notifications ensure that users are promptly informed of potential hazards, reducing the risk of appliance damage or fire accidents. This layer bridges the gap between the system and its users, making energy management intuitive and responsive.

4. Communication Layer

The communication layer ensures remote monitoring and control, leveraging the IoT capabilities of the ESP32 microcontroller. Data collected by sensors is transmitted in real time to a cloud server or IoT platform. The mobile application provides users with live updates, control functionality, and alert notifications. The system employs secure communication protocols to ensure data integrity and privacy. Historical data storage enables energy consumption analysis, trend identification, and informed decision-making for energy optimization. This layer enhances scalability, allowing the integration of additional appliances, smart meters, or renewable energy sources.

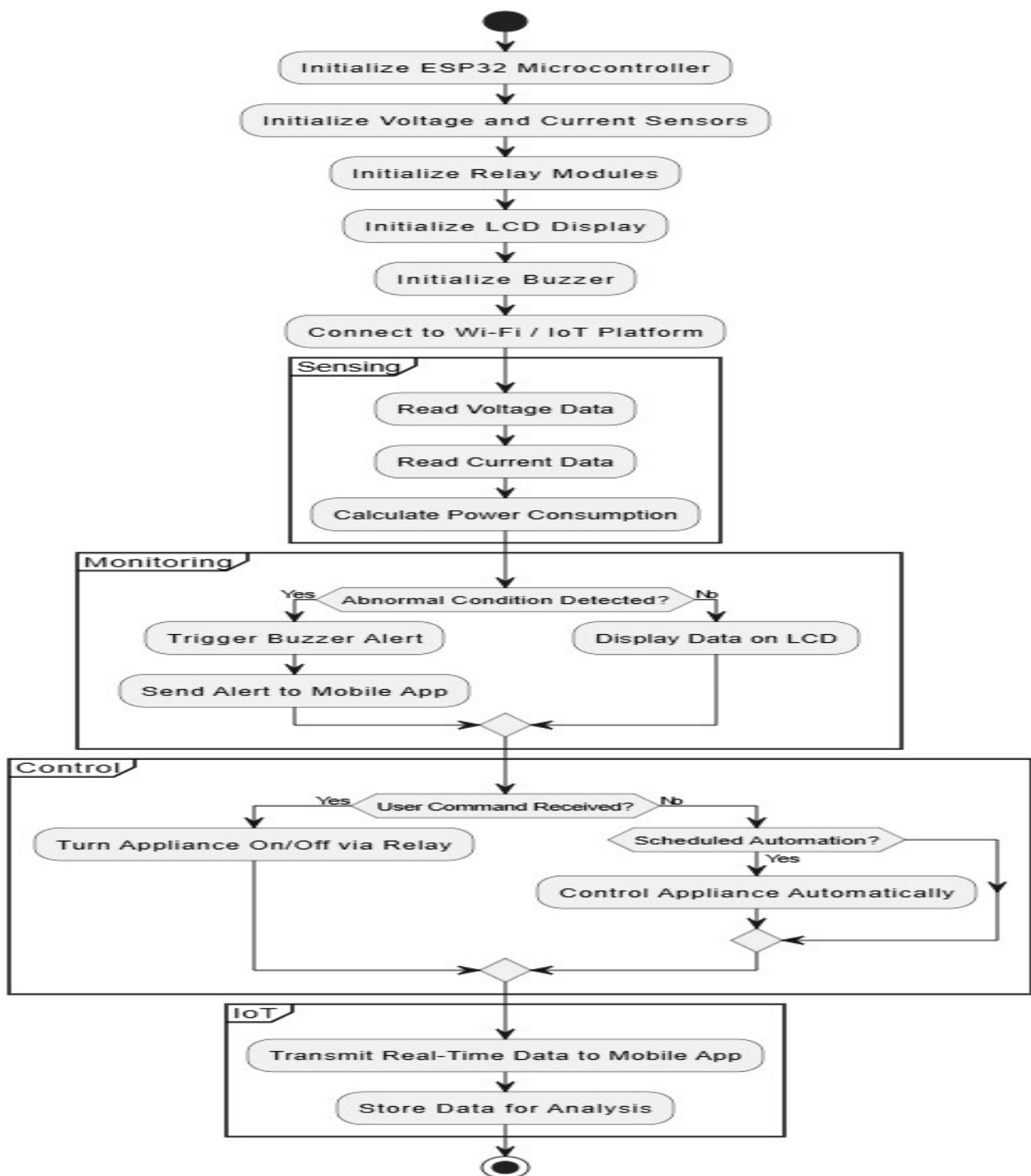
Integration and Workflow

The system architecture integrates all layers into a seamless workflow. Sensor data is captured and processed by the ESP32, triggering relay actions, updating the LCD display, generating alerts, and transmitting information to the mobile application. This closed-loop architecture ensures synchronized monitoring, control, and feedback. Each layer communicates with the others in real time, maintaining efficiency, accuracy, and safety throughout the system.

Summary

In conclusion, the system architecture of the Smart Home Electricity Monitoring and Control System is modular, scalable, and user-centric. The integration of sensing, control, display, and communication layers provides a comprehensive framework that enables real-time energy monitoring, automated appliance management, proactive safety measures, and remote access. This architecture forms the backbone of a modern smart home energy solution that is reliable, efficient, and adaptable to future technological enhancements.

FLOW CHART:



CHAPTER 6

6.MODULES

1. ESP32 Microcontroller Unit (200 words)

The ESP32 microcontroller functions as the central processing unit of the system, orchestrating data acquisition, control, and communication processes. It is selected due to its high-speed dual-core processor, built-in Wi-Fi, low power consumption, and extensive compatibility with sensor and relay modules. The ESP32 receives real-time input from voltage and current sensors, calculates power consumption, and executes programmed logic to control appliance operation. It serves as the interface between the sensing layer, control layer, display layer, and IoT communication layer. In addition, it manages thresholds for abnormal conditions, triggering buzzer alerts and sending notifications to the mobile application. The microcontroller's versatility enables the integration of automated scheduling, remote control commands, and data logging for analysis. By centralizing all computational and control functions, the ESP32 ensures synchronized and efficient system operation. Its embedded firmware is responsible for reading sensor inputs, calculating instantaneous electrical parameters, controlling relays, updating the LCD, and facilitating secure data transmission to the cloud platform. Overall, the ESP32 microcontroller forms the backbone of the system, enabling intelligent, real-time, and user-friendly energy management within modern smart homes.

2. Voltage and Current Sensing Module (200 words)

The voltage and current sensing module is responsible for capturing accurate real-time electrical parameters essential for monitoring and control. Voltage sensors measure the potential difference across the supply line, while current sensors determine the flow of electricity through individual appliances or circuits. The data collected from these sensors is transmitted to the ESP32 microcontroller for processing, enabling the calculation of instantaneous power consumption. These sensors are calibrated to ensure precision and

reliability under varying load conditions. The sensing module plays a critical role in detecting abnormal conditions such as overcurrent, voltage spikes, and potential short circuits. By providing real-time input, it allows the system to execute preventive measures through the relay control module and trigger buzzer alerts. Moreover, the sensing module contributes to energy optimization by supplying detailed data for analysis, enabling users to identify high-energy-consuming devices and adjust usage patterns. The integration of voltage and current sensors ensures that the system not only monitors overall consumption but also evaluates the operational status of individual appliances. This module forms the foundation of a responsive, intelligent, and energy-efficient smart home management system.

3. Relay Control Module (200 words)

The relay control module enables the automated and remote operation of electrical appliances, forming the active control layer of the system. Relays act as electrically operated switches, allowing the ESP32 microcontroller to turn devices such as lights, fans, and other household equipment on or off based on sensor readings, scheduled automation, or remote commands from the mobile application. This module ensures that appliances operate efficiently, minimizing unnecessary energy consumption and enhancing user convenience. The relay control module is designed to handle standard household voltage and current ratings, ensuring safe and reliable operation. In case of abnormal conditions detected by the sensing module, the relays can disconnect appliances automatically to prevent damage and reduce the risk of electrical hazards. By integrating with the ESP32, this module supports both manual remote control and intelligent automation, enabling scheduled or conditional operations. Furthermore, the relay control module allows scalability, as additional appliances or circuits can be incorporated without significant system redesign. It serves as a critical interface between intelligent decision-making and physical appliance management, transforming real-time monitoring into actionable control, thereby optimizing energy usage and ensuring a safe, modern smart home environment.

4. LCD Display Module (200 words)

The LCD display module provides a visual interface for users to observe real-time electrical

data and system status. It presents voltage, current, power consumption, and appliance operation information, enabling users to monitor energy usage patterns and make informed decisions. By offering immediate feedback, the LCD facilitates energy awareness and allows the detection of anomalies before they escalate into critical hazards. The display is directly interfaced with the ESP32 microcontroller, which updates information continuously, ensuring that readings reflect real-time electrical conditions. This module complements remote monitoring by providing localized visualization, which is especially useful during initial setup or in environments where mobile access is unavailable. Additionally, the LCD supports user-friendly interaction by showing system alerts, operational schedules, and maintenance reminders. The integration of the display module enhances the system's transparency, allowing users to understand the correlation between appliance operation and energy consumption. By making energy data accessible and interpretable, the LCD display module contributes to improved energy efficiency, user convenience, and proactive household management. It is a vital component for bridging automated control and human interaction, ensuring that the smart home system remains intuitive, informative, and actionable.

5. Buzzer Alert Module (200 words)

The buzzer alert module serves as an auditory notification system, providing immediate warnings when the system detects abnormal electrical conditions. It is triggered when the voltage or current sensors identify overcurrent, voltage fluctuations, or short circuits, ensuring rapid awareness of potential hazards. By delivering an audible alert, the buzzer enhances safety, allowing users to take corrective action before damage occurs to appliances or property. The module is directly controlled by the ESP32 microcontroller, which evaluates sensor readings and activates the buzzer based on predefined threshold limits. This real-time alert mechanism is particularly useful in scenarios where users are not continuously monitoring the LCD display or mobile application. Additionally, the buzzer works in conjunction with the IoT module, ensuring that notifications are also transmitted remotely via the mobile application. Its integration into the system ensures redundancy in safety alerts, reinforcing household protection measures. By providing instantaneous auditory feedback,

the buzzer alert module strengthens the reliability and effectiveness of the smart home electricity management system, supporting both preventive and proactive energy and safety practices in modern residential environments.

6. IoT Connectivity and Mobile Application Module (200 words)

The IoT connectivity and mobile application module enables remote monitoring, control, and management of the smart home electricity system. By leveraging the ESP32 microcontroller's Wi-Fi capability, real-time electrical data from sensors is transmitted to a cloud platform accessible through a mobile application. Users can view voltage, current, and power consumption for individual appliances, receive alerts for abnormal conditions, and control devices remotely from any location. This module enhances user convenience by providing a centralized interface for energy management, reducing dependency on physical presence. It also supports historical data logging, allowing trend analysis, energy optimization, and identification of high-energy-consuming appliances. Secure communication protocols ensure the integrity and privacy of transmitted data. Furthermore, this module facilitates system scalability, enabling integration with additional smart devices, renewable energy sources, or advanced energy management algorithms in the future. By combining remote accessibility, real-time monitoring, and control, the IoT module transforms the traditional home electricity system into an intelligent, interactive, and user-friendly solution. It ensures that energy efficiency, appliance management, and safety are maintained even when the user is offsite, making it a cornerstone of modern smart home technology.

7. Cost Prediction Module

The cost prediction module uses a machine learning algorithm to estimate the monthly electricity bill based on collected energy consumption data. The system processes parameters such as voltage, current, and usage duration to calculate power consumption. Using historical data, the model predicts the expected monthly cost, helping users monitor and optimize their electricity usage.

CHAPTER 7

CODING AND TESTING

7.1 CODING

The coding phase of the Smart Home Electricity Monitoring and Control System forms the backbone of its intelligent operation, integrating sensor data acquisition, appliance control, alert generation, display updates, and IoT communication into a cohesive software framework. The ESP32 microcontroller serves as the primary platform for coding due to its high computational capacity, dual-core processor, built-in Wi-Fi, and compatibility with various sensor and actuator modules. Programming is primarily carried out using the Arduino IDE with C/C++ language extensions, which allows seamless integration with libraries for sensors, relays, LCD displays, and IoT protocols.

The first component of coding involves sensor data acquisition. Voltage and current sensors are read using analog input pins, and raw readings are converted into meaningful electrical parameters such as voltage, current, and power using calibration coefficients and mathematical formulas. The software continuously monitors these parameters in real time, enabling precise measurement of energy consumption and detection of anomalies. Threshold values for overcurrent, voltage fluctuation, and power limits are defined within the program, which allows the system to respond immediately when deviations occur.

The relay control module is implemented through digital output pins controlled by the microcontroller. The coding logic determines whether appliances should be turned on or off based on sensor input, user commands received via IoT, or pre-programmed schedules. This allows automated energy management and ensures that appliances are only operational when necessary, reducing power wastage. The software also updates the LCD display at regular intervals, providing real-time information such as voltage, current, power consumption, and appliance status.

For safety, the coding integrates a buzzer alert routine that is triggered if sensor readings exceed defined thresholds. This immediate feedback mechanism ensures user awareness of abnormal conditions, preventing potential hazards. Parallel to local alerts, the coding includes IoT integration, allowing real-time data transmission to a mobile application using protocols such as MQTT or HTTP. The software handles secure connection, data formatting, and periodic updates, ensuring that users can monitor and control appliances remotely.

Additionally, the code is modular and scalable, allowing future integration of new sensors, relays, or smart home devices without significant modifications. Error-handling routines are included to manage communication failures, sensor disconnections, or invalid readings, enhancing system robustness. Overall, the coding phase translates the conceptual design into actionable software that enables real-time sensing, intelligent control, remote monitoring, and user interaction, forming the functional core of the smart home electricity management system.

7.2. TEST PROCEDURE

SYSTEM TESTING

Testing is a critical phase in the development of the Smart Home Electricity Monitoring and Control System, ensuring that the integrated hardware and software components operate correctly, reliably, and safely. The primary objective of testing is to validate the system's functionality, performance, safety, and interoperability under real-world conditions. Testing is performed in structured stages, including functional testing, performance testing, safety testing, integration testing, and user acceptance testing, each focusing on specific aspects of the system.

1. FUNCTIONAL TESTING

Functional testing aims to verify that each module performs its intended operation accurately. The voltage and current sensing module is tested by applying known loads to the circuit and

comparing the sensor readings with standard measurement instruments. The ESP32 microcontroller processes these readings and calculates instantaneous power consumption, which is cross-verified against manual calculations to ensure accuracy. The relay control module is tested by sending on/off commands from both the microcontroller and the mobile application to confirm proper switching of appliances such as lights and fans. The LCD display module is evaluated to ensure that it presents real-time electrical data, including voltage, current, and power consumption, in an understandable format. The buzzer alert module is tested by simulating abnormal conditions such as overcurrent, voltage fluctuations, or short circuits to verify that alerts are triggered promptly. Functional testing ensures that each component operates according to design specifications and that users can interact with the system effectively.

2. PERFORMANCE TESTING

Performance testing evaluates the system's responsiveness, stability, and real-time processing capabilities. The sensor data acquisition process is monitored for update frequency and accuracy, ensuring that readings reflect real-time electrical conditions. The ESP32 microcontroller's ability to process data and execute control commands under different load conditions is assessed to determine processing efficiency. The relay response time is measured to ensure appliances switch within milliseconds of a control command. IoT communication is also evaluated by measuring the latency of data transmission to the mobile application and the responsiveness of remote control commands. Stress testing is performed by simultaneously operating multiple appliances at maximum load to examine the system's performance under extreme conditions. Any delays, data loss, or miscommunication are identified and corrected to maintain real-time monitoring and control.

3. SAFETY TESTING

Safety testing ensures the system can protect both users and appliances under abnormal

conditions. The voltage and current thresholds defined in the ESP32 firmware are validated by simulating overcurrent, short circuits, and voltage surges. When these conditions occur, the buzzer alert module is expected to trigger immediately, notifying users of the potential hazard. Simultaneously, the relay module is programmed to disconnect the affected appliance to prevent further damage. Remote notifications via the mobile application are also tested to confirm users are alerted in real time. Safety testing validates the reliability of both local and remote preventive measures, ensuring that the system mitigates risks such as appliance damage, fire hazards, and electrical shocks.

4. INTEGRATION TESTING

Integration testing evaluates the interoperability of all system modules working together as a cohesive unit. The ESP32 microcontroller receives sensor data, processes it, updates the LCD, controls appliances through relays, triggers buzzer alerts, and communicates with the mobile application simultaneously. Test scenarios include normal operation, scheduled appliance control, remote control commands, and abnormal conditions. Integration testing ensures that data flow between modules is synchronized, responses are timely, and system stability is maintained. Any conflicts, delays, or malfunctioning interactions are identified and corrected. This phase confirms that all hardware and software components function harmoniously, providing a seamless user experience.

5. IOT AND REMOTE ACCESS TESTING

The IoT module is tested for data transmission reliability, latency, and security. Sensor data is sent to a cloud server and visualized on a mobile application. Users are tested for the ability to receive real-time updates, remotely switch appliances on or off, and respond to alerts. Network stability and data integrity are evaluated under different Wi-Fi conditions, including intermittent connectivity. Security testing is also performed to ensure authentication protocols protect against unauthorized access. This ensures that remote monitoring and

control are both effective and secure.

6. USER ACCEPTANCE TESTING

User acceptance testing evaluates the system's usability, effectiveness, and user satisfaction. Test users interact with the mobile application and LCD interface to monitor electricity consumption, control appliances, and respond to alerts. Feedback is collected regarding clarity of display, responsiveness of controls, and ease of operation. Adjustments are made based on user input to improve the interface and overall system experience.

7. DATA VALIDATION AND ENERGY OPTIMIZATION TESTING

The system stores historical energy consumption data to enable analysis and energy optimization. Data logging is tested for accuracy, completeness, and retrieval capability. Historical trends are analyzed to identify high-energy-consuming appliances and peak consumption periods. Recommendations for scheduling and optimized operation are verified against actual energy savings achieved.

8. SUMMARY OF TESTING RESULTS

Testing confirms that the system accurately monitors electrical parameters, controls appliances efficiently, provides real-time alerts, and enables secure remote access. Functional, performance, safety, integration, IoT, and user acceptance tests collectively validate the system's reliability, responsiveness, and usability. Stress testing ensures stability under maximum load, while historical data analysis verifies energy optimization capability. The comprehensive testing process demonstrates that the Smart Home Electricity Monitoring and Control System meets its design objectives, offering accurate monitoring, automated appliance control, enhanced electrical safety, and user-friendly IoT-enabled remote access. The system is validated as a robust and scalable solution suitable for modern smart home energy management.

CHAPTER 8

8. Screenshots

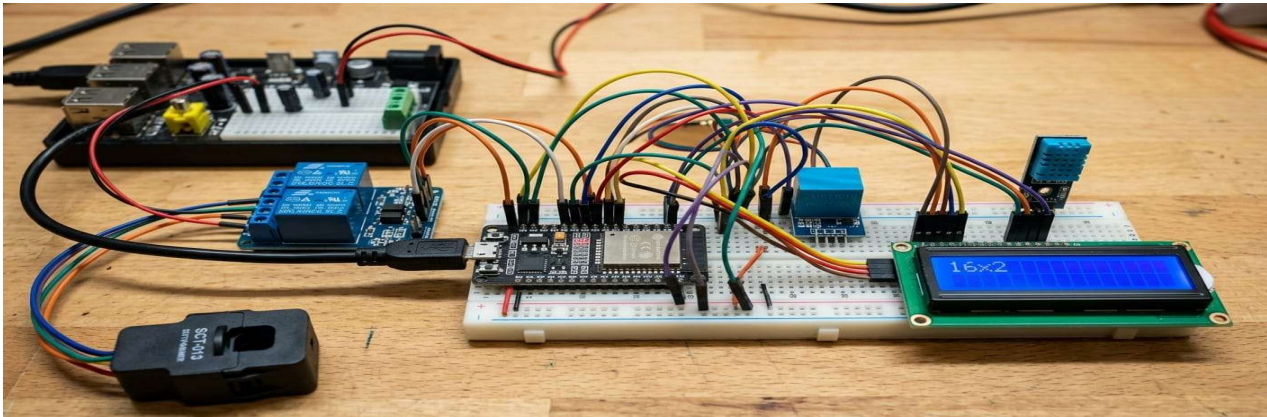


Figure 1 Hardware Setup of Smart Home Monitoring System

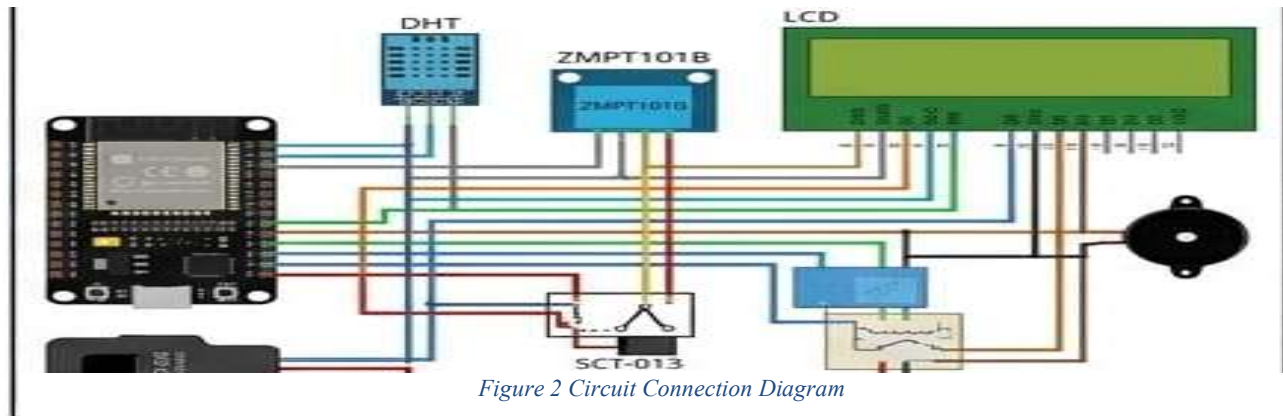


Figure 2 Circuit Connection Diagram



Figure 3 Real-Time Sensor Data Display



Figure 4 Mobile Application Monitoring Interface

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